

Algebra II with Trigonometry

Chapter 7.2

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Here is a complete breakdown and study guide based on the provided document.

1. Summary of the Document

This document is a guided notes worksheet for an Algebra 2 Trigonometry/Honors class covering the **Addition and Subtraction Formulas** (also known as sum and difference formulas) for sine, cosine, and tangent. It outlines how to use these formulas to find the exact values of non-standard angles, simplify expressions, and prove trigonometric identities. Additionally, it provides specific techniques for evaluating expressions that contain inverse trigonometric functions, and for combining linear combinations of sine and cosine ($A \sin x + B \cos x$) into a single phase-shifted sine function.

2. Core Definitions, Theorems, and Formulas

Addition and Subtraction Formulas:

- **Sine:**
 - $\sin(s + t) = \sin s \cos t + \cos s \sin t$
 - $\sin(s - t) = \sin s \cos t - \cos s \sin t$
- **Cosine:**
 - $\cos(s + t) = \cos s \cos t - \sin s \sin t$
 - $\cos(s - t) = \cos s \cos t + \sin s \sin t$
- **Tangent:**
 - $\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$

- $\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$

Evaluating Inverse Trigonometric Expressions:

- An expression involving an inverse trigonometric function, such as $\cos^{-1} x$, represents an **angle**.

Expressions of the Form $A \sin x + B \cos x$:

- Any linear combination of sine and cosine can be rewritten as a single sine wave using the formula:

$$A \sin x + B \cos x = k \sin(x + \phi)$$

where:

- $k = \sqrt{A^2 + B^2}$ (the amplitude)
- The angle ϕ is found using: $\cos \phi = \frac{A}{\sqrt{A^2 + B^2}}$ and $\sin \phi = \frac{B}{\sqrt{A^2 + B^2}}$

3. Worked Study Guide

Concept 1: Finding Exact Values of Non-Standard Angles (Based on Example 1) You can evaluate angles that aren't on the standard unit circle by splitting them into the sum or difference of angles that *are* on the unit circle (like 30° , 45° , 60°).

- **Step 1:** Identify two standard angles that add or subtract to the target angle. For example, to find $\cos(\frac{17\pi}{12})$, you can rewrite $\frac{17\pi}{12}$ as $\frac{8\pi}{12} + \frac{9\pi}{12}$, which simplifies to $\frac{2\pi}{3} + \frac{3\pi}{4}$.
- **Step 2:** Apply the appropriate sum/difference formula.
 $\cos(\frac{2\pi}{3} + \frac{3\pi}{4}) = \cos(\frac{2\pi}{3}) \cos(\frac{3\pi}{4}) - \sin(\frac{2\pi}{3}) \sin(\frac{3\pi}{4})$.
- **Step 3:** Substitute the known unit circle values and simplify.

Concept 2: Simplifying Expressions (Based on Example 2) If you are given a long expression that fits the right side of an addition/subtraction formula, you can collapse it back down to a single trig function.

- *Step-by-step:* Looking at $\cos 23^\circ \cos 67^\circ - \sin 23^\circ \sin 67^\circ$, notice this matches the structure $\cos s \cos t - \sin s \sin t$, which is the formula for $\cos(s + t)$.
- Substitute $s = 23^\circ$ and $t = 67^\circ$ to get $\cos(23^\circ + 67^\circ) = \cos(90^\circ)$. Since $\cos(90^\circ) = 0$, the whole expression simplifies to exactly 0.

Concept 3: Proving Identities (Based on Example 3) To prove an identity, start with the more complex side and use the sum/difference formulas to expand it, then simplify until it matches the other side.

- *Example:* Prove $\cos(x - \frac{\pi}{2}) = \sin x$.
- *Step 1:* Expand the left side using the cosine subtraction formula: $\cos x \cos(\frac{\pi}{2}) + \sin x \sin(\frac{\pi}{2})$.
- *Step 2:* Substitute known values. We know $\cos(\frac{\pi}{2}) = 0$ and $\sin(\frac{\pi}{2}) = 1$.
- *Step 3:* Simplify: $\cos x(0) + \sin x(1) = \sin x$. The identity is proven!

Concept 4: Working with Inverse Trig Functions (Based on Example 5) Remember the rule: an inverse trig function represents an *angle*.

- *Example:* Evaluate $\sin(\cos^{-1}(\frac{1}{2}) + \tan^{-1}(1))$.
- *Step 1:* Assign variables to the angles. Let $\alpha = \cos^{-1}(\frac{1}{2})$ and $\beta = \tan^{-1}(1)$.
- *Step 2:* Use the unit circle to find these exact angles (if they are standard). Here, $\alpha = \frac{\pi}{3}$ and $\beta = \frac{\pi}{4}$. (If they weren't standard, you would draw right reference triangles to find the missing side lengths).
- *Step 3:* You are now evaluating $\sin(\frac{\pi}{3} + \frac{\pi}{4})$. Use the sine addition formula: $\sin(\frac{\pi}{3}) \cos(\frac{\pi}{4}) + \cos(\frac{\pi}{3}) \sin(\frac{\pi}{4})$.
- *Step 4:* Substitute and solve: $(\frac{\sqrt{3}}{2})(\frac{\sqrt{2}}{2}) + (\frac{1}{2})(\frac{\sqrt{2}}{2}) = \frac{\sqrt{6} + \sqrt{2}}{4}$.

Concept 5: Writing $A \sin x + B \cos x$ as a Single Sine Function (Based on Example 7)

- *Example:* Write $-\sqrt{3} \sin x + \cos x$ in terms of sine only.
- *Step 1:* Identify $A = -\sqrt{3}$ and $B = 1$.
- *Step 2:* Calculate

$$k = \sqrt{A^2 + B^2} = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4} = 2.$$

- **Step 3:** Find angle ϕ . We know $\cos \phi = \frac{-\sqrt{3}}{2}$ and $\sin \phi = \frac{1}{2}$. This means ϕ is in Quadrant II, specifically $\phi = \frac{5\pi}{6}$.
- **Step 4:** Write the final equation: $k \sin(x + \phi) \rightarrow 2 \sin(x + \frac{5\pi}{6})$.

4. Practice Problems

Problem 1: Use an addition or subtraction formula to find the exact value of $\sin(15^\circ)$. (*Hint: $15^\circ = 45^\circ - 30^\circ$*) **Problem 2:** Write the expression $\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$ as a trigonometric function of one number, and find its exact value. **Problem 3:** Prove the identity: $\sin(x + \pi) = -\sin x$. **Problem 4:** Suppose $\sin \theta = \frac{3}{5}$ (θ is in Quadrant I) and $\cos \phi = \frac{5}{13}$ (ϕ is in Quadrant I). Evaluate $\sin(\theta + \phi)$. (*Hint: use the Pythagorean theorem to find the missing trig values first*). **Problem 5:** Write the expression $3 \sin x + 4 \cos x$ in terms of sine only. Leave the phase shift angle ϕ in the form of an inverse trigonometric function.

Answers to Practice Problems: 1. $\frac{\sqrt{6}-\sqrt{2}}{4}$ *Process:*

$$\sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right).$$

2. $\sin(60^\circ) = \frac{\sqrt{3}}{2}$ *Process:* Matches $\sin(s + t)$. $\sin(40^\circ + 20^\circ) = \sin(60^\circ)$.

3. **Proof:** Expand the left side: $\sin x \cos \pi + \cos x \sin \pi$. Substitute unit circle values for π : $\sin x(-1) + \cos x(0) = -\sin x$. This matches the right side. 4. $\frac{63}{65}$ *Process:* If $\sin \theta = \frac{3}{5}$, drawing a triangle shows $\cos \theta = \frac{4}{5}$. If $\cos \phi = \frac{5}{13}$, a triangle shows $\sin \phi = \frac{12}{13}$.

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi = \left(\frac{3}{5}\right)\left(\frac{5}{13}\right) + \left(\frac{4}{5}\right)\left(\frac{12}{13}\right) = \frac{15}{65} + \frac{48}{65} = \frac{63}{65}$$

5. $5 \sin(x + \cos^{-1}(\frac{3}{5}))$ or $5 \sin(x + \sin^{-1}(\frac{4}{5}))$ *Process:* $A = 3, B = 4$.

$k = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. We need an angle ϕ where $\cos \phi = \frac{3}{5}$ and $\sin \phi = \frac{4}{5}$. Therefore, $\phi = \cos^{-1}(\frac{3}{5})$.

5. Assumptions and Ambiguities Resolved

- **OCR Errors Handled:** The OCR (text recognition) output heavily scrambled the mathematical formatting, fractions, minus signs, and omitted π symbols (e.g., parsing $\frac{17\pi}{12}$ as $17\backslash\pi 12$). I relied exclusively on

the visual screenshots provided to ensure all formulas, variables, and examples were captured correctly.

- **Fill-in-the-Blank Context:** In Section I, the text leaves a blank before "angle". By reading the context cues of the original image, I inferred the completed sentence should read "remember that an expression like $\cos^{-1} x$ represents an **angle**."
- **Example 2a Resolution:** The expression in example 2a in the image contains a tiny typo/ambiguity in standard spacing ($\cos 23^\circ \cos 67^\circ - \sin 23^\circ \sin 67^\circ$). By correctly applying the $\cos(s + t)$ rule, I resolved that the problem is asking to show this equates to $\cos(90^\circ)$, which is simply 0.